

Research Notes on Attention mechanism neural network

Attention mechanism neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

Predecessors For many years, sequence modelling and generation was done by using plain recurrent neural networks (RNNs). A well-cited early example was the Elman network (1990). In theory, the information from one token can propagate arbitrarily far down the sequence, but in practice the vanishing-gradient problem leaves the model's state at the end of a long sentence without precise, extractable information about preceding tokens. A key breakthrough was LSTM (originally described in a 1995 technical report and formally published in 1997), an RNN that introduced gating mechanisms to mitigate the vanishing gradient problem, allowing efficient learning of long-sequence modelling. One key architectural element was the use of multiplicative gating units, in which the outputs of some neurons modulate the outputs of others. These multiplicative units are conceptually distinct from the additive attention mechanism later introduced for sequence-to-sequence models. Neural networks using multiplicative units were later called sigma-pi networks or higher-order networks. LSTM became the standard architecture for long sequence modelling until the 2017 publication of transformers. However, LSTM still used sequential processing, like most other RNNs. Specifically, RNNs operate one token at a time from first to last; they cannot operate in parallel over all tokens in a sequence. Modern transformers overcome this problem, but unlike RNNs, they require computation time that is quadratic in the size of the context window. The linearly scaling fast weight controller (1992) learns to compute a weight matrix for further processing depending on the input. One of its two networks has "fast weights" or "dynamic links" (1981). A slow neural network learns by gradient descent to generate keys and values for computing the weight changes of the fast neural network which computes answers to queries. This was later shown to be equivalent to the unnormalized linear transformer.

>> Section 2

Scaled dot-product attention and self-attention The use of the scaled dot-product attention and self-attention mechanism instead of a recurrent neural network or long short-term memory (which rely on recurrence instead) allows for better performance as described in the following paragraph. The paper described the scaled dot-product attention as follows:

>> Section 3

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$
$$\{ \displaystyle PE_{\{pos\}, 2i} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \}$$

>> Section 4

$$Attention(Q, K, V) := \text{softmax}\left(\frac{Q \times K^T}{d_k}\right) \times V$$
$$\{ \displaystyle Attention(Q, K, V) := \text{softmax}\left(\frac{Q \times K^T}{d_k}\right) \times V \}$$

>> Section 5

Background The authors of the paper are Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan Gomez, Łukasz Kaiser, and Illia Polosukhin. All eight authors were "equal contributors" to the paper; the listed order was randomized (according to the paper itself). After the paper, each of the authors left Google to join other companies or to found startups. The paper's title is a reference to the song "All You Need Is Love" by the Beatles. The name "Transformer" was picked because Jakob Uszkoreit, one of the paper's authors, liked the sound of that word. An early design document was titled "Transformers: Iterative Self-Attention and Processing for Various Tasks", and included an illustration of six characters from the Transformers franchise. The team was named Team Transformer.

>> Section 6

The idea of encoder-decoder sequence transduction had been developed in the early 2010s; commonly cited as the originators that produced seq2seq are two concurrently published papers from 2014. A 380M-parameter model for machine translation uses two long short-term memories (LSTM). Its architecture consists of two parts. The encoder is an LSTM that takes in a sequence of tokens and turns it into a vector. The decoder is another LSTM that converts the vector into a sequence of tokens. Similarly, another 130M-parameter model used gated recurrent units (GRU) instead of LSTM. Later research showed that GRUs are neither better nor worse than LSTMs for seq2seq. These early seq2seq models had no attention mechanism, and the state vector is accessible only after the last word of the source text was processed. Although in theory such a vector retains the information about the whole original sentence, in practice the information is poorly preserved. This is because the input is processed sequentially by one recurrent network into a fixed-size output vector, which is then processed by another recurrent network into an output. If the input is long, then the output vector would not be able to contain all relevant information, degrading the output. As evidence, reversing the input sentence improved seq2seq translation. The RNN search model introduced an attention mechanism to seq2seq for

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machine translation to solve the bottleneck problem (of the fixed-size output vector), allowing the model to process long-distance dependencies more easily. The name is because it "emulates searching through a source sentence during decoding a translation". The relative performances were compared between global (that of RNN search) and local (sliding window) attention model architectures for machine translation, finding that mixed attention had higher quality than global attention, while local attention reduced translation time. In 2016, Google Translate was revamped to Google Neural Machine Translation, which replaced the previous model based on statistical machine translation. The new model was a seq2seq model where the encoder and the decoder were both 8 layers of bidirectional LSTM. It took nine months to develop, and it outperformed the statistical approach, which took ten years to develop.

>> Section 7

Seq2seq models with attention (including self-attention) still suffered from the same issue with recurrent networks, which is that they are hard to parallelize, which prevented them from being accelerated on GPUs. In 2016, decomposable attention applied a self-attention mechanism to feedforward networks, which are easy to parallelize, and achieved SOTA result in textual entailment with an order of magnitude fewer parameters than LSTMs. One of its authors, Jakob Uszkoreit, suspected that attention without recurrence would be sufficient for language translation, thus the title "attention is all you need". That hypothesis was against conventional wisdom at the time, and even his father Hans Uszkoreit, a well-known computational linguist, was skeptical. In the same year, self-attention (called intra-attention or intra-sentence attention) was proposed for LSTMs. On 2017-06-12, the original (100M-parameter) encoder-decoder transformer model was published in the "Attention is all you need" paper. At the time, the focus of the research was on improving seq2seq for machine translation, by removing its recurrence to process all tokens in parallel, but preserving its dot-product attention mechanism to keep its text processing performance. This led to the introduction of a multi-head attention model that was easier to parallelize due to the use of independent heads and the lack of recurrence. Its parallelizability was an important factor to its widespread use in large neural networks.